
IncidentResponseAI

Hindsight-Powered On-Call SRE Agent

An AI SRE agent that remembers previous incidents and uses that knowledge to recommend better, faster, evidence-based resolutions.

Team	Members	Hackathon
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 THE PROBLEM

Production Incidents Repeat. Knowledge Gets Lost.



Scattered knowledge

Incidents, logs, and runbooks live in separate places.



AI forgets context

Generic tools give one-off answers with no history.



Same fixes, again

Teams repeat the same troubleshooting from scratch.



Minutes are lost

Outage context is buried when it matters most.

Why Existing Approaches Fall Short

Each solves part of the problem — but only IncidentResponseAI combines persistent memory with real-time intelligence.

Generic AI

Stateless

Runbooks

Manual

Tickets

Unstructured

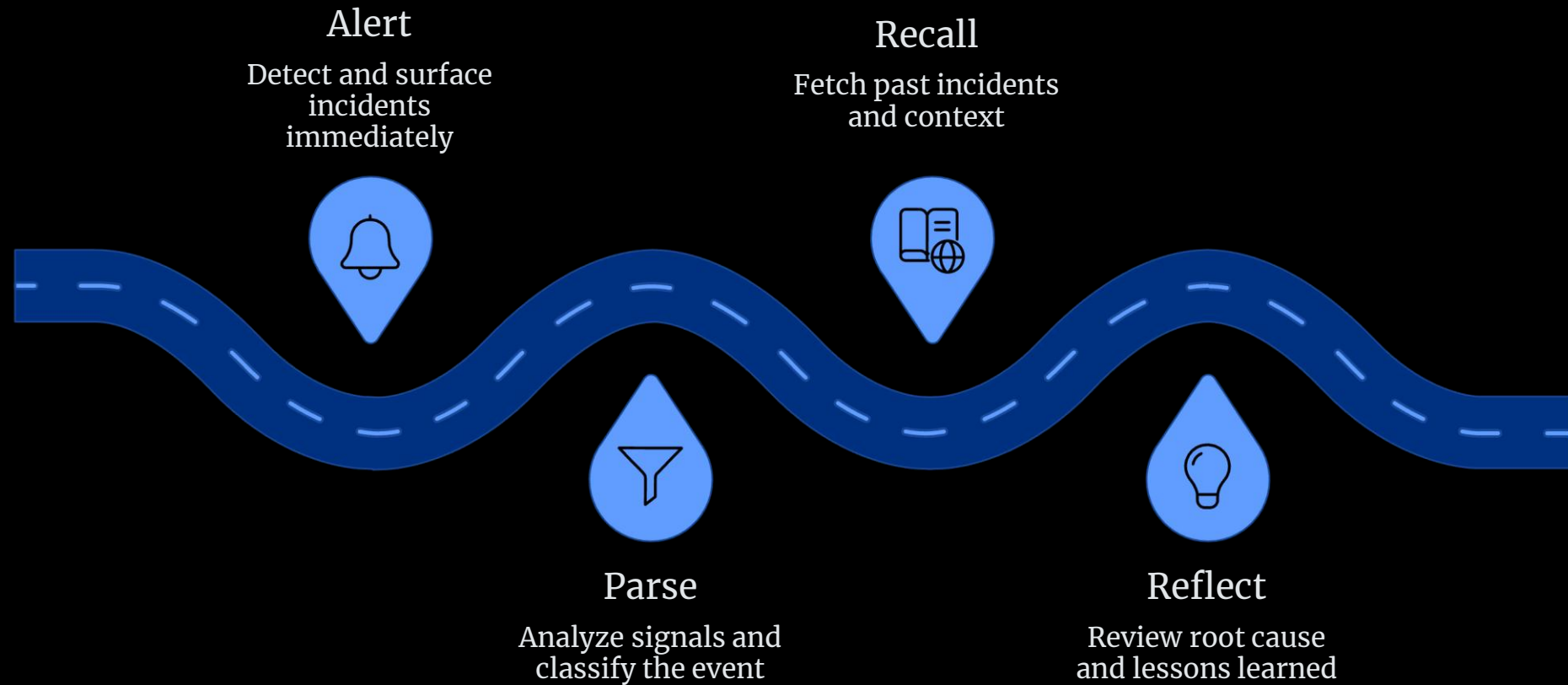
One-Shot

No learning

❏ “IncidentResponseAI makes persistent memory the core of the workflow.”

Our Solution: IncidentResponseAI

A continuous cycle that transforms every incident into reusable operational knowledge. **Hindsight** ensures that no resolution is ever forgotten.



Hindsight turns every resolved incident into reusable operational knowledge — the loop compounds with every cycle.

How Hindsight Memory Powers It

Three memory operations keep SRE intelligence learning from every incident.



retain()

Stores incident context and resolution details.



recall()

Finds similar incidents and proven fixes fast.



reflect()

Turns past evidence into better recommendations.

FEATURE SET

Key Features



Incident Memory

Stores incident context and resolutions.



Evidence-Based Runbooks

Suggests proven fixes from prior cases.



Historical Recall

Finds similar past incidents fast.

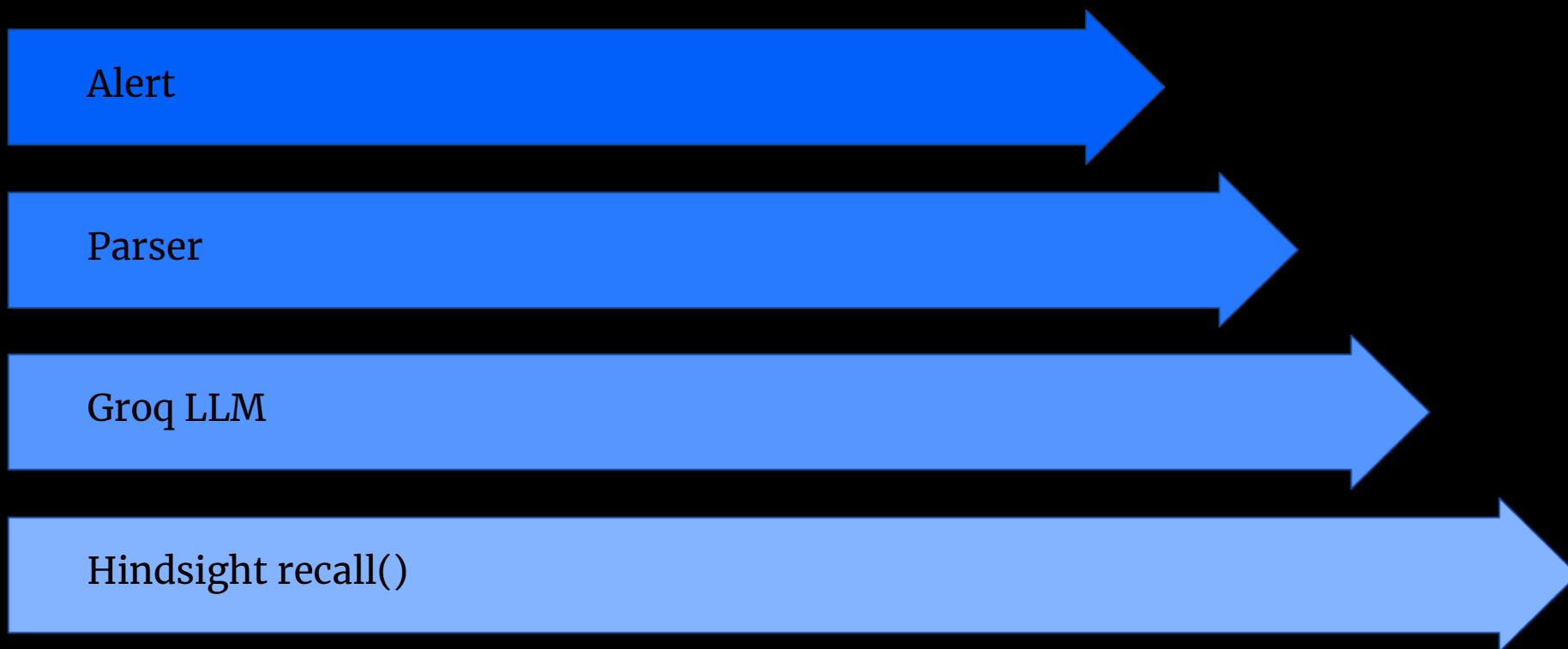


Continuous Learning

Improves with every resolved incident.

Technical Pipeline

How IncidentResponseAI processes incidents end-to-end — from alert ingestion to memory-enriched resolution and knowledge retention.



The pipeline is designed for real-time operation — **Groq** provides sub-second LLM inference while **Hindsight** memory operations (**recall()**, **reflect()**, **retain()**) run in parallel, ensuring no latency is added to the critical resolution path.

Architecture & Technology Stack

A lean stack for rapid MVP development and realistic demonstration.

System Components

Frontend

Streamlit

AI Engine

Groq

Memory

Hindsight

Backend

Python

Data

Synthetic incidents

Data Flow



SRE Alert

Streamlit UI

Agent Core

Runbook +
Resolution

Alert → UI → Analysis → Resolution

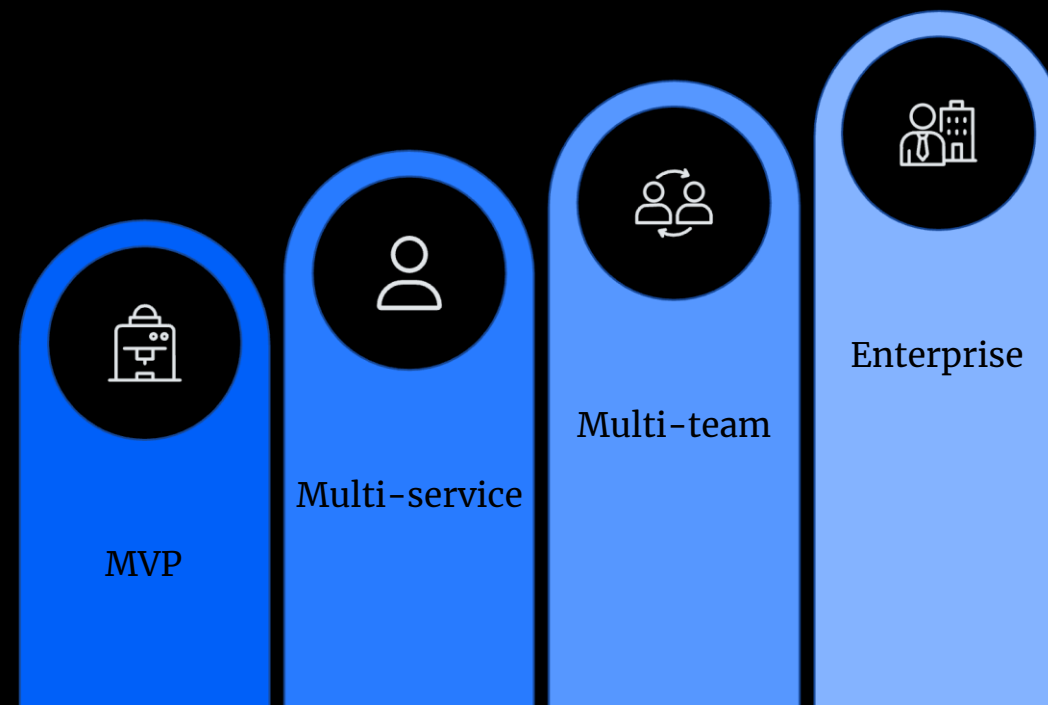
Feasibility & Scalability

MVP Feasibility

- Hindsight memory
Persistent memory ready to use.
- Groq inference
Fast real-time recommendations.
- Python + Streamlit
Rapid build and demo.
- Synthetic data
Repeatable, credible scenarios.

❗ *Post-MVP: monitoring, Slack, Jira, PagerDuty, CI/CD.*

Scalability Path



Expands from a single demo service to organization-wide incident memory.

☆ REAL-WORLD IMPACT

Real-World Impact



Faster Resolution

Surface relevant historical fixes faster.



Reduced Repetition

Reuse proven solutions instead of starting over.



Consistency at Scale

Every engineer follows the same evidence-based playbook.



Knowledge Preservation

Critical expertise stays with the system.



Faster Onboarding

Help new responders leverage existing incident knowledge.

Future scope: real-time monitoring, Slack/Jira/PagerDuty automation, post-mortems, and human-approved remediation.

Every incident becomes memory. Every memory improves the next response.